

A General Semantics for Analogical Reasoning

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Analogical reasoning

- Pivotal element in human cognitive processes
- Central concept underlying vector representations in machine learning
- Why a formal semantics and logic of analogy
 - ▶ To endow artificial agents with analogical reasoning capacities
 - ▶ To verify whether analogical reasoning performed by LLMs is correct

Analogueal proportion (AP)

- It traces back to Aristotle
- Analogy as the equality of two ratios (logos) with at least four members

Views of AP

$$A:B :: C:D \approx \text{“}A \text{ is to } B \text{ as } C \text{ is to } D\text{”}$$

⇒ Functional view [Hofstadter et al., 1995]:

$$\text{“}\exists f \in \mathbf{F} \text{ such that } f(A) = B \text{ and } f(C) = D\text{”}$$

⇒ Relational view [Gentner, 1983]:

$$\text{“}\exists \mathcal{R} \in \mathbf{R} \text{ such that } A\mathcal{R}B \text{ and } C\mathcal{R}D\text{”}$$

⇒ Differential view [Miclet & Prade, 2009]:

$$\text{“}A \text{ differs from } B \text{ as } C \text{ differs from } D \text{ and} \\ B \text{ differs from } A \text{ as } D \text{ differs from } C\text{”}$$

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Generality: relational > functional > differential

Views of AP

– $2:4 :: -3:9$

(with $f = x \mapsto x^2$)

RoadRunner:WileCoyote :: Tweety:Sylvester

(with $\mathcal{R} = \textit{predatedBy}$)

Basic postulates for AP

(refl) $A:B :: A:B$

(sym) If $A:B :: C:D$ then $C:D :: A:B$

(cperm) If $A:B :: C:D$ then $A:C :: B:D$

Derived properties:

- $A : A :: B : B$ (*identity*)
- $A : B :: C : D \Rightarrow D : B :: C : A$ (*extreme permutation*)
- $A : B :: C : D \Rightarrow B : A :: D : C$ (*internal reversal*)
- $A : B :: C : D \Rightarrow D : C :: B : A$ (*complete reversal*)

Other postulates for AP

(unic) If $A:B :: C:D$ and $A:B :: C:E$ then $D = E$

(trans) If $A:B :: C:D$ and $C:D :: E:F$ then $A:B :: E:F$

Contribution of the paper

- Semantics for the *differential view* of AP recently proposed in [Herzig, Lorini, Prade, 2024]
 - ▶ Terms of an AP are **logical formulas** denoting facts
- We introduce a model-theoretic semantics for the *relational view* of AP between **logical formulas**

Model theory

Signature: countable set of atomic propositions $\mathbb{P}$

Definition (Relational model)

A relational model is a tuple $M = (W, \Theta, \mathcal{V})$ where:

- W is a non-empty set of possible worlds;
- $\Theta \subseteq \wp(\wp(W) \times \wp(W))$ is a set of relations between facts;
- $\mathcal{V} : W \rightarrow 2^{\mathbb{P}}$ is a valuation function.

The class of relational models is denoted by $\mathbf{M}$.

Example

$$king \stackrel{\text{def}}{=} male \wedge royal$$

$$queen \stackrel{\text{def}}{=} \neg male \wedge royal$$

- $W = \{w_1, w_2, w_3, w_4\}$
- $\mathcal{V}(w_1) = \{male\}$
- $\mathcal{V}(w_2) = \{royal\}$
- $\mathcal{V}(w_3) = \{male, royal\}$
- $\mathcal{V}(w_4) = \{\}$
- $\Theta = \{\mathcal{R}_{marriedTo}, \mathcal{R}_{subjectTo}\}$ with
 - ▶ $\mathcal{R}_{marriedTo} = \{(\{w_2\}, \{w_3\}), (\{w_3\}, \{w_2\})\}$
 - ▶ $\mathcal{R}_{subjectTo} = \{(\{w_1, w_4\}, \{w_2\}), (\{w_1, w_4\}, \{w_3\})\}$

Modal language

Language $\mathcal{L}$:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \leadsto \varphi \mid \Box\varphi$$

with p ranging over $\mathbb{P}$

Modal language

Language $\mathcal{L}$:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \rightsquigarrow \varphi \mid \Box\varphi$$

with p ranging over $\mathbb{P}$

$\Box\varphi$: “ φ is true for all relations”

$\varphi \rightsquigarrow \psi$: “the fact φ is related to the fact ψ ”

Dual modality:

$$\Diamond\varphi \stackrel{\text{def}}{=} \neg\Box\neg\varphi$$

Semantic interpretation

Let $M = (W, \Theta, \mathcal{V}) \in \mathbf{M}$, $\mathcal{R} \in \Theta$, $w \in W$. Then:

$$\begin{aligned}(M, \mathcal{R}, w) \models p &\iff p \in \mathcal{V}(w) \\(M, \mathcal{R}, w) \models \neg\varphi &\iff (M, \mathcal{R}, w) \not\models \varphi \\(M, \mathcal{R}, w) \models \varphi \wedge \psi &\iff (M, \mathcal{R}, w) \models \varphi \text{ and } (M, \mathcal{R}, w) \models \psi \\(M, \mathcal{R}, w) \models \varphi \rightsquigarrow \psi &\iff \llbracket \varphi \rrbracket_M \mathcal{R} \llbracket \psi \rrbracket_M \\(M, \mathcal{R}, w) \models \Box\varphi &\iff \forall \mathcal{R}' \in \Theta, (M, \mathcal{R}', w) \models \varphi\end{aligned}$$

with $\llbracket \varphi \rrbracket_M = \{v \in W : \forall \mathcal{R} \in \Theta, (M, \mathcal{R}, v) \models \varphi\}$

Semantic interpretation

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$(M, \mathcal{R}, w) \models \varphi_1 : \varphi_2 :: \varphi_3 : \varphi_4$ iff $\exists \mathcal{R}' \in \Theta$ such that $\llbracket \varphi_1 \rrbracket_M \mathcal{R}' \llbracket \varphi_2 \rrbracket_M$ and $\llbracket \varphi_3 \rrbracket_M \mathcal{R}' \llbracket \varphi_4 \rrbracket_M$.

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Fact

We have:

$$(M, \mathcal{R}, w) \models \varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \text{ iff } (M, \mathcal{R}, w) \models \Diamond((\varphi_1 \rightsquigarrow \varphi_2) \wedge (\varphi_3 \rightsquigarrow \varphi_4)).$$

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$$\varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \stackrel{\text{def}}{=} \Diamond((\varphi_1 \rightsquigarrow \varphi_2) \wedge (\varphi_3 \rightsquigarrow \varphi_4))$$

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$\forall w \in W, \forall \mathcal{R} \in \Theta:$

$$(M, \mathcal{R}, w) \models (queen:king :: king:queen) \wedge (\neg royal:queen :: \neg royal:king)$$

How to validate properties of AP

(refl) $\varphi_1:\varphi_2 :: \varphi_1:\varphi_2$

(sym) If $\varphi_1:\varphi_2 :: \varphi_3:\varphi_4$ then $\varphi_3:\varphi_4 :: \varphi_1:\varphi_2$

(cperm) If $\varphi_1:\varphi_2 :: \varphi_3:\varphi_4$ then $\varphi_1:\varphi_3 :: \varphi_2:\varphi_4$

(unic) If $\varphi_1:\varphi_2 :: \varphi_3:\varphi_4$ and $\varphi_1:\varphi_2 :: \varphi_3:\varphi_5$ then $\varphi_4 = \varphi_5$

(trans) If $\varphi_1:\varphi_2 :: \varphi_3:\varphi_4$ and $\varphi_3:\varphi_4 :: \varphi_5:\varphi_6$ then $\varphi_1:\varphi_2 :: \varphi_5:\varphi_6$

How to validate properties of AP

The postulate (*sym*) holds by construction. Indeed, we have the following validity:

$$\models_{\mathbf{M}} \varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \rightarrow \varphi_3:\varphi_4 :: \varphi_1:\varphi_2 \quad (1)$$

How to validate properties of AP

Constraints on relational models:

(C1) $\forall X_1, X_2 \in \wp(W), \exists \mathcal{R} \in \Theta$ such that $X_1 \mathcal{R} X_2$

(C2) $\forall \mathcal{R} \in \Theta, \forall X_1, X_2, X_3, X_4 \in \wp(W)$, (if $(X_1 \mathcal{R} X_2$ and $X_3 \mathcal{R} X_4)$ then $(\exists \mathcal{R}' \in \Theta$ such that $X_1 \mathcal{R}' X_3$ and $X_2 \mathcal{R}' X_4)$)

(C3) $\forall \mathcal{R}, \mathcal{R}' \in \Theta$, if $\mathcal{R} \neq \mathcal{R}'$ then $\mathcal{R} \cap \mathcal{R}' = \emptyset$ (disjointness)

(C4) $\forall \mathcal{R} \in \Theta, \forall X_1, X_2, X_3 \in \wp(W)$, if $X_1 \mathcal{R} X_2$ and $X_1 \mathcal{R} X_3$ then $X_2 = X_3$ (functionality)

How to validate properties of AP

$$\text{(Reflexivity)} \models_{\mathbf{M}_{\{C1\}}} \varphi_1:\varphi_2 :: \varphi_1:\varphi_2 \quad (2)$$

$$\text{(Central perm.)} \models_{\mathbf{M}_{\{C2\}}} \varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \rightarrow \varphi_1:\varphi_3 :: \varphi_2:\varphi_4 \quad (3)$$

$$\text{(Unicity)} \models_{\mathbf{M}_{\{C3\}}} (\varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \wedge \varphi_3:\varphi_4 :: \varphi_5:\varphi_6) \rightarrow \varphi_1:\varphi_2 :: \varphi_5:\varphi_6 \quad (4)$$

$$\text{(Transitivity)} \models_{\mathbf{M}_{\{C4\}}} (\varphi_1:\varphi_2 :: \varphi_3:\varphi_4 \wedge \varphi_1:\varphi_2 :: \varphi_3:\varphi_5) \rightarrow \Box(\varphi_4 \leftrightarrow \varphi_5) \quad (5)$$

Research agenda: proof-theory

$\Rightarrow$ Sound and complete axiomatization for the language $\mathcal{L}$ for each model class $\mathbf{M}_C$ with $C \subseteq \{C1, C2, C3, C4\}$

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Some notable validities:

$$\models_{\mathbf{M}_{\{C1\}}} \Diamond(\varphi \rightsquigarrow \psi) \tag{6}$$

$$\models_{\mathbf{M}_{\{C2\}}} ((\varphi \rightsquigarrow \psi) \wedge (\chi \rightsquigarrow \gamma)) \rightarrow \Diamond((\varphi \rightsquigarrow \chi) \wedge (\psi \rightsquigarrow \gamma)) \tag{7}$$

$$\models_{\mathbf{M}_{\{C3\}}} ((\varphi \rightsquigarrow \psi) \wedge \chi) \rightarrow \Box((\varphi \rightsquigarrow \psi) \rightarrow \chi) \tag{8}$$

$$\models_{\mathbf{M}_{\{C4\}}} ((\varphi \rightsquigarrow \psi) \wedge (\varphi \rightsquigarrow \chi)) \rightarrow \Box(\psi \leftrightarrow \chi) \tag{9}$$

Research agenda: extended language

Extension with world-quantifier modality $\blacksquare$:

$$(M, \mathcal{R}, w) \models \blacksquare\varphi \iff \forall v \in W, (M, \mathcal{R}, v) \models \varphi$$

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Resulting logic: instance of the modal product $S5^n$

Research agenda: complexity analysis and decision procedures

- Complexity of satisfiability checking for base language and its extension
- Tableau-based decision procedures

Thank you for your attention!